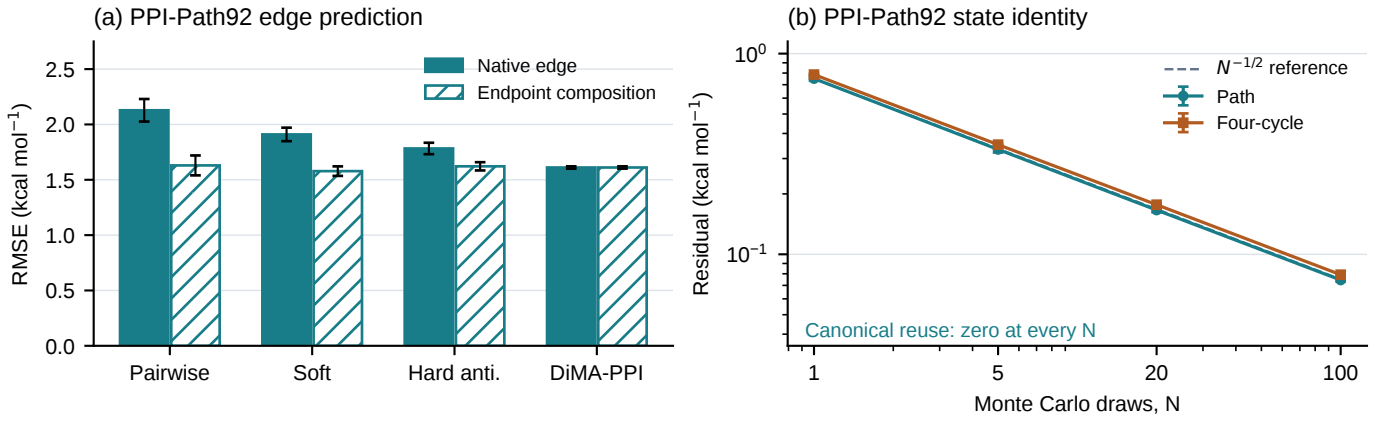
(a) PPI-Path92 edge prediction
RMSE (kcal mol^{-1})
2.5
2.0
1.5
1.0
0.5
0.0
Native edge
Endpoint composition
Pairwise
Soft
Hard anti.
DiMA-PPI
(b) PPI-Path92 state identity
Residual (kcal mol^{-1})
10^{0}
10^{-1}
$N^{-1/2}$ reference
Path
Four-cycle
Canonical reuse: zero at every N
1
5
20
100
Monte Carlo draws, N